



Department of Economics
Johns Hopkins University
Baltimore, MD 21218
dedwar65@jh.edu

November 11, 2025

Hiring Committee
Responsible AI Institute (RAI)
145 Broadway
Cambridge, MA 02142

Dear Hiring Committee,

I am writing to express my interest in the Research Scientist – Ethics and Society position at the Responsible AI Institute (RAI). I am a Ph.D. candidate in the Department of Economics at Johns Hopkins University (advisors: Christopher D. Carroll, Nicholas W. Papageorge, and Stelios Fourakis), and I expect to receive my degree in May 2026. My primary areas of specialization are macroeconomic modeling, wealth and income dynamics, and computational economics.

My research agenda focuses on the ability of macroeconomic models to generate empirically realistic distributions of wealth and income over the life cycle. A key driver of that work is modeling heterogeneity in the rate of return on assets, and understanding how return dispersion contributes to portfolio accumulation across households. These themes connect naturally to RAI's mission to ensure that data, algorithms, and technology are designed and deployed in ways that are equitable, transparent, and socially responsible.

In my job-market paper, I calibrate a life-cycle heterogeneous-agent model to match wealth moments from the Survey of Consumer Finances by allowing households to differ in their portfolio returns. The model helps explain observed wealth concentration and return dispersion. In a related project using the Health and Retirement Study, I examine how trust influences both income growth and asset returns — finding a hump-shaped relationship for returns, and characterizing the distribution of the persistent component of net-worth returns. These experiences have equipped me to: (i) frame research strategies and design modeling approaches; (ii) execute econometric and simulation analysis with large microdata sets; and (iii) translate quantitative findings into actionable insights for academic, policy, and industry audiences.

What draws me most to RAI is its commitment to integrating rigorous analytics with ethical reflection and societal impact. My background in quantitative modeling and empirical analysis complements RAI's interdisciplinary work at the intersection of economics, technology, and governance. I am particularly motivated by the opportunity to develop frameworks that connect data transparency, algorithmic accountability, and social equity — principles that parallel my own research philosophy of grounding theory in measurable, real-world behavior.

I believe I would be a strong fit for the Research Scientist – Ethics and Society position because:

1. My computational and econometric expertise (Python, R, and Stata) allows me to build and validate complex models while maintaining transparency and replicability.

2. My research directly engages with questions of inequality, trust, and decision-making — themes central to understanding the societal impacts of technology and algorithmic systems.
3. I bring a collaborative mindset and communication skills well-suited to RAI's interdisciplinary environment, where research informs ethical standards and policy applications.

I am deeply inspired by RAI's mission to align technological innovation with human values and fairness. I would welcome the opportunity to contribute my analytical, modeling, and empirical expertise to your team's research and policy initiatives.

Thank you for considering my application.

Sincerely,

Decory Edwards